$$\lim_{x\to 0^+} x\ln x = 0$$

1. $\ln x := \int_1^x \frac{1}{t}dt \exists, \forall x>0$

proof: ① $\because \frac{1}{t} \downarrow$ on $[a,b]$, $0<a\le b$

$\therefore \int_a^b \frac{1}{t}dt \ \exists$

② $\because \forall x>0, \int_1^x \frac{1}{t}dt \exists$.

$\therefore \ln(x)$ is defined for $x>0$

2. $\lim_{x\to\infty} \frac{\ln x}{x} = 0$

proof: $\because t > t^{\frac{1}{2}}$, if $t>1$

$\therefore$ for $x>1$

$$0 \le \ln x = \int_1^x \frac{1}{t}dt$$
$$< \int_1^x \frac{1}{t^{\frac{1}{2}}}dt$$
$$= 2(\sqrt{x}-1)$$
$$< 2\sqrt{x}$$

$\therefore 0 \le \frac{\ln x}{x} < \frac{2}{\sqrt{x}}$

$\therefore 0 \le \lim_{x\to\infty} \frac{\ln x}{x} \le \lim_{x\to\infty} \frac{2}{\sqrt{x}} = 0$ □

3. $\lim_{x\to 0^+} x\ln x = \lim_{t\to\infty} -\frac{\ln t}{t} = 0.$